

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Applied Business Analytics Practical

Practical – I(Self Study)

Roll No.: T016	Name: Himani Malankar
Class:TYIT	Batch:1
Date of Assignment: 22/06/2026	Date/Time of Submission: 22/06/2026

Use Netflix.csv

Q.1 Write a Python program to:

- Find the Top 10 Genres.
- Find the Bottom 10 Genres.
- Identify the most popular Genre.
- Identify the least popular Genre.
- Display the result using a bar chart.

CODE:

```
#Task1
# Import required libraries
import pandas as pd
import matplotlib.pyplot as plt

# Load the dataset
df = pd.read_csv("Netflix.csv")

# Count the number of titles in each genre
genre_count = df["genres"].value_counts()

# Display Top 10 Genres
print("Top 10 Genres")
print(genre_count.head(10))

# Display Bottom 10 Genres
print("\nBottom 10 Genres")
print(genre_count.tail(10))

# Display Most Popular Genre
print("\nMost Popular Genre")
print(genre_count.idxmax(), "-", genre_count.max())

# Display Least Popular Genre
print("\nLeast Popular Genre")
print(genre_count.idxmin(), "-", genre_count.min())

# Plot Top 10 Genres
plt.figure(figsize=(10,5))
genre_count.head(10).plot(kind="bar", color="skyblue")

plt.title("Top 10 Genres")
plt.xlabel("Genre")
plt.ylabel("Number of Titles")

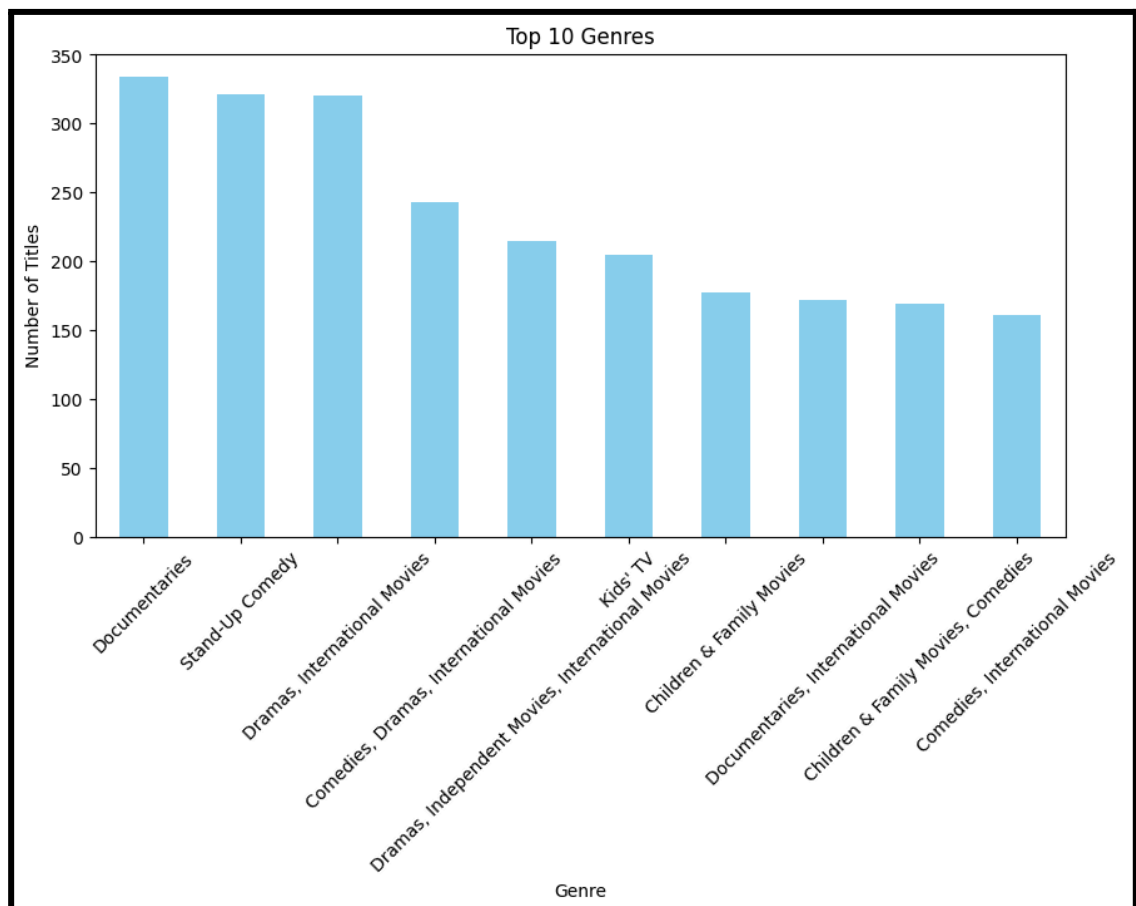
plt.xticks(rotation=45)

plt.show()
```

OUTPUT:

Top 10 Genres	Bottom 10 Genres
genres	genres
Documentaries 334	International Movies, Romantic Movies, Thrillers 1
Stand-Up Comedy 321	Comedies, Cult Movies, Independent Movies 1
Dramas, International Movies 320	Sports Movies 1
Comedies, Dramas, International Movies 243	Children & Family Movies, Comedies, Faith & Spirituality 1
Dramas, Independent Movies, International Movies 215	Action & Adventure, Horror Movies, Independent Movies 1
Kids' TV 205	British TV Shows, Docuseries, TV Comedies 1
Children & Family Movies 177	Horror Movies, International Movies, Sci-Fi & Fantasy 1
Documentaries, International Movies 172	TV Comedies, TV Sci-Fi & Fantasy, Teen TV Shows 1
Children & Family Movies, Comedies 169	Action & Adventure, Classic Movies, International Movies 1
Comedies, International Movies 161	Stand-Up Comedy & Talk Shows, TV Mysteries, TV Sci-Fi & Fantasy 1
Name: count, dtype: int64	Name: count, dtype: int64

Most Popular Genre Documentaries - 334 Least Popular Genre TV Comedies, TV Horror, TV Thrillers - 1



Q.2 Write a Python program to find:

- The oldest movie.
- The newest movie.
- The oldest TV Show.
- The newest TV Show.

Display all details of these titles.

CODE:

```
# Separate Movies and TV Shows
movies = df[df["type"]=="Movie"]
tvshows = df[df["type"]=="TV Show"]

# Oldest Movie
print("Oldest Movie")
print(movies[movies["release_year"]==movies["release_year"].min()])

# Newest Movie
print("\nNewest Movie")
print(movies[movies["release_year"]==movies["release_year"].max()])

# Oldest TV Show
print("\nOldest TV Show")
print(tvshows[tvshows["release_year"]==tvshows["release_year"].min()])

# Newest TV Show
print("\nNewest TV Show")
print(tvshows[tvshows["release_year"]==tvshows["release_year"].max()])
```

OUTPUT:

Oldest Movie

show_id	type	title	director \
4402	s4961	Movie	Prelude to War Frank Capra
5688	s6118	Movie	The Battle of Midway John Ford

	cast	country	date_added	release_year \
4402	NaN	United States	31-Mar-17	1942
5688	Henry Fonda, Jane Darwell	United States	31-Mar-17	1942

	rating	duration	genres \
4402	TV-14	52	Classic Movies, Documentaries
5688	TV-14	18	Classic Movies, Documentaries

Newest Movie

show_id	type	title \
319	s1286	Movie
397	s1356	Movie
574	s1515	Movie
589	s1529	Movie
4125	s4711	Movie
4562	s5104	Movie

5470	What happens after we die? This docuseries exp...
6087	A boy suddenly orphaned fights his parents' ki...
6337	Hosts David Spade, Fortune Feimster and London...
7767	Follow LA's wildly wealthy Asian and Asian Ame...

	duration	genres \
319	85	Children & Family Movies, Comedies
397	98	Stand-Up Comedy
574	90	Documentaries
589	14	Documentaries
4125	116	Action & Adventure
4562	76	Documentaries, International Movies, Music & M...
5401	97	Comedies, Dramas, International Movies
6302	54	Documentaries
6914	96	Dramas, International Movies
7281	102	Comedies, International Movies
7301	32	Documentaries
7384	102	Dramas, International Movies, LGBTQ Movies

	description
319	On the eve of his 21st birthday, an adored pri...
397	In this extended cut of his 2018 special, Chri...
574	A cheap, powerful drug emerges during a recess...
589	A fascinating character. Exquisite sets. A wig...
4125	In the near future, a drone pilot sent into a ...
4562	The irrepressible Ratones Paranoicos, Argentin...
5401	Entrenched in a midlife crisis, Aziz seeks sol...
6302	They've built a movement out of minimalism. Lo...
6914	When her estranged mother falls into a coma, a...
7281	With the peak of his career long behind him, a...
7301	In this delightful short documentary, an Itali...
7384	Singing and dreaming together, a talented sing...

Oldest TV Show											
show_id	type	title director cast \									
4298	s4868	TV Show	Pioneers: First Women Filmmakers*							NaN	NaN
country	date_added	release_year	rating	duration	genres \						
4298	NaN	30-Dec-18	1925	TV-14	1	TV Shows					
description											
4298 This collection restores films from women who ...											

Oldest TV Show										
show_id	type	title director cast \								
4298	s4868	TV Show	Pioneers: First Women Filmmakers*						NaN	NaN
country	date_added	release_year	rating	duration	genres \					
4298	NaN	30-Dec-18	1925	TV-14	1	TV Shows				
description										
4298	This collection restores films from women who ...									

Newest TV Show			
show_id	type	title \	
250	s1223	TV Show	Carmen Sandiego
492	s1441	TV Show	Cobra Kai
869	s1781	TV Show	Disenchantment
962	s1865	TV Show	Dream Home Makeover
1477	s2328	TV Show	Gabby's Dollhouse
1860	s2673	TV Show	Headspace Guide to Meditation
1937	s2742	TV Show	Hilda
1950	s2754	TV Show	History of Swear Words
2245	s3019	TV Show	Inside the World's Toughest Prisons
3138	s3823	TV Show	Lupin
3431	s4087	TV Show	Mighty Little Bheem: Kite Festival

Q.3 Write a Python program to:

- Find the country with the highest Netflix content.
- Display the Top 10 Countries.
- Display the Bottom 10 Countries.
- Display the result using a horizontal bar chart.

CODE:

```
# Count content by country
country_count = df["country"].dropna().value_counts()

# Display country with highest content
print("Country with Highest Netflix Content")
print(country_count.head(1))

# Display Top 10 Countries
print("\nTop 10 Countries")
print(country_count.head(10))

# Display Bottom 10 Countries
print("\nBottom 10 Countries")
print(country_count.tail(10))

# Plot Horizontal Bar Chart
plt.figure(figsize=(10,6))
country_count.head(10).sort_values().plot(kind="barh", color="orange")
plt.title("Top 10 Countries by Netflix Content")
plt.xlabel("Number of Titles")
plt.ylabel("Country")

plt.show()
```

OUTPUT:

```
... Country with Highest Netflix Content
country
United States    2555
Name: count, dtype: int64
```

Top 10 Countries

country

```
United States    2555
India            923
United Kingdom   397
Japan            226
South Korea      183
Canada           177
Spain            134
France           115
Egypt            101
Mexico           100
```

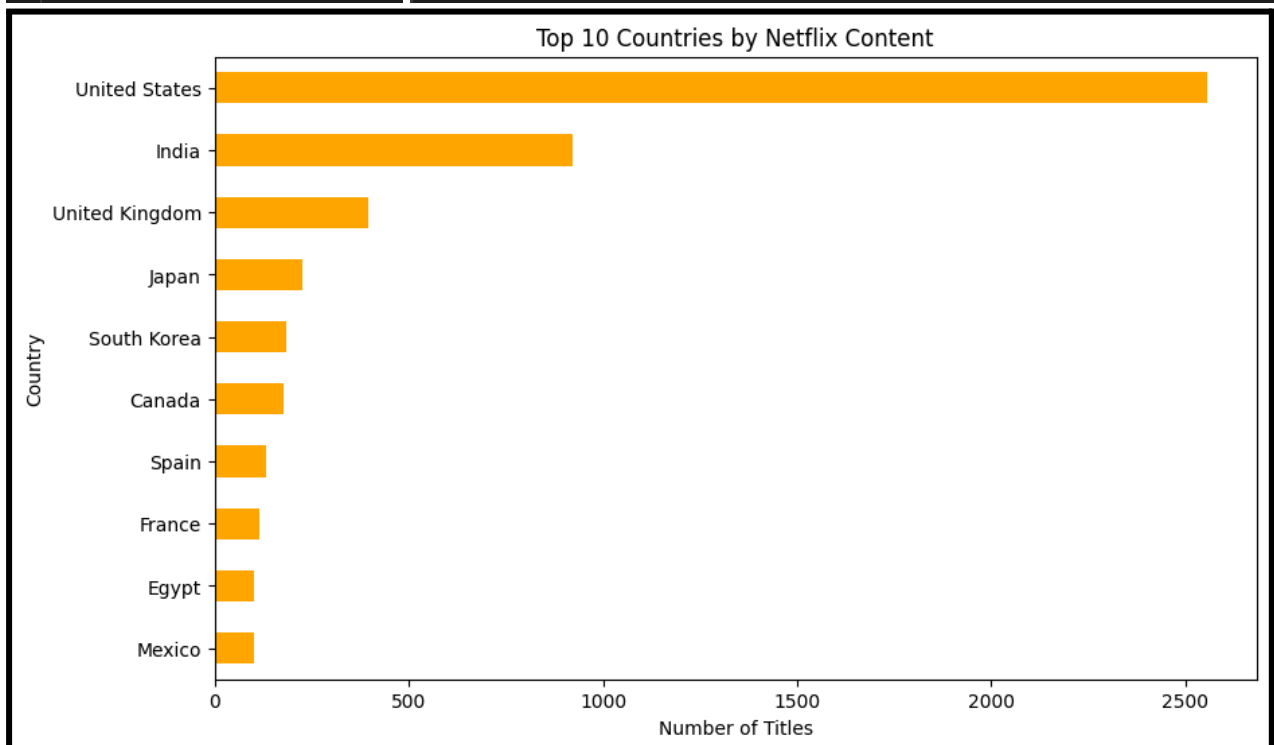
Name: count, dtype: int64

Bottom 10 Countries

country

```
Poland, West Germany    1
Italy, United Kingdom, France    1
China, Taiwan           1
France, United Kingdom, India    1
United States, Canada, France    1
Argentina, United States, Mexico    1
Canada, Germany, France, United States    1
Soviet Union, India        1
Taiwan, China, France, United States    1
France, Belgium, Spain     1
```

Name: count, dtype: int64



Q.4 Write a Python program to:

- Find the director with the highest number of Movies.
- Find the director with the highest number of TV Shows.
- Display the Top 10 Directors.
- Plot a bar chart of the Top 10 Directors.

CODE:

```
# Movies only
movie_director = df[df["type"]=="Movie"]["director"].dropna().value_counts()

# TV Shows only
tv_director = df[df["type"]=="TV Show"]["director"].dropna().value_counts()

# Display Director with Highest Movies
print("Director with Highest Number of Movies")
print(movie_director.head(1))

# Display Director with Highest TV Shows
print("\nDirector with Highest Number of TV Shows")
print(tv_director.head(1))

# Display Top 10 Directors
director_count = df["director"].dropna().value_counts()
print("\nTop 10 Directors")
print(director_count.head(10))

# Plot Bar Chart
plt.figure(figsize=(10,5))
director_count.head(10).plot(kind="bar", color="green")
plt.title("Top 10 Directors")
plt.xlabel("Director")
plt.ylabel("Number of Titles")
plt.xticks(rotation=45)

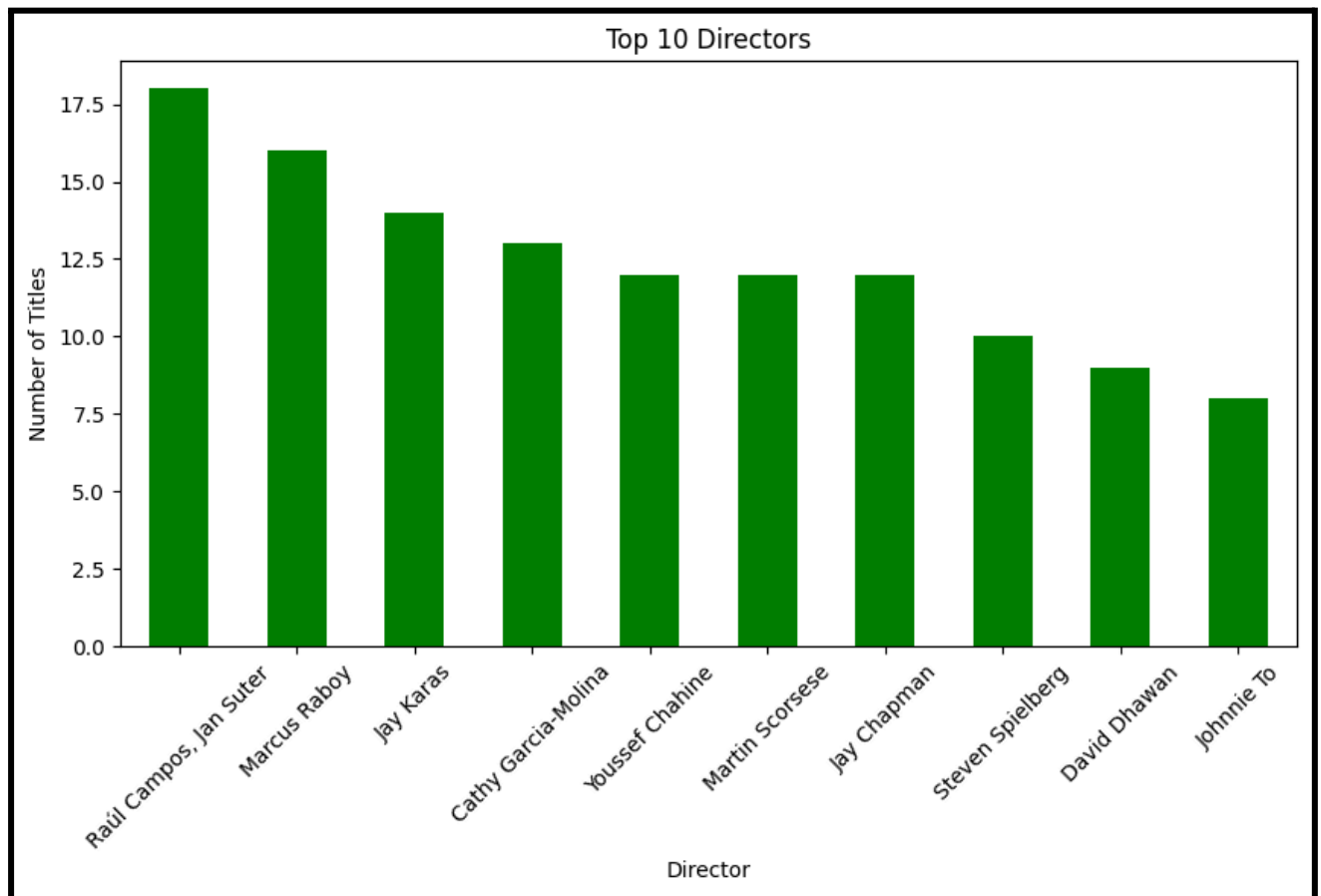
plt.show()
```

OUTPUT:

```
... Director with Highest Number of Movies
director
Raúl Campos, Jan Suter    18
Name: count, dtype: int64
```

```
Director with Highest Number of TV Shows
director
Alastair Fothergill    3
Name: count, dtype: int64
```

```
Top 10 Directors
director
Raúl Campos, Jan Suter    18
Marcus Raboy             16
Jay Karas                 14
Cathy Garcia-Molina      13
Youssef Chahine           12
Martin Scorsese           12
Jay Chapman               12
Steven Spielberg          10
David Dhawan              9
Johnnie To                8
Name: count, dtype: int64
```



Q.5 Compare Netflix releases:

- Before 2020.
- During 2020.
- After 2020.
- Display the result using a bar chart or line chart.

CODE:

```
# Count releases
before_2020 = df[df["release_year"] < 2020].shape[0]
during_2020 = df[df["release_year"] == 2020].shape[0]
after_2020 = df[df["release_year"] > 2020].shape[0]

# Create DataFrame
release_data = pd.DataFrame({
    "Period": ["Before 2020", "During 2020", "After 2020"],
    "Count": [before_2020, during_2020, after_2020]
})
print(release_data)

# Plot Bar Chart
plt.figure(figsize=(7,5))
plt.bar(release_data["Period"], release_data["Count"], color=["blue", "orange", "green"])
plt.title("Netflix Releases Comparison")
plt.xlabel("Period")
plt.ylabel("Number of Titles")

plt.show()
```

OUTPUT:

```
...
   Period  Count
0 Before 2020  6888
1 During 2020   868
2 After 2020    31
```

